

WORKFORCE ADMINISTRATION SOLUTION(DEV)

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Team ID	LTVIP2025TMID29622
Project name	Workforce administration solution(DEV)
Maximum marks	

Chapter – 4

Project Design

4.1 - Existing System (Problem Solution Fit)

1. Target Customer / User Segment

- Primary Users: HR managers, operations managers, team leads, payroll administrators
- Secondary Users: Employees, contractors, compliance officers
- Company Profile: Medium to large-sized businesses with 50–5,000 employees, often in industries like healthcare, logistics, manufacturing, or IT services.

2. Problem Statement

→ Core Problem: Administrative workforce management processes are manual, fragmented, or reliant on legacy systems.

→ Consequences:

- ◆Time wasted on repetitive tasks
- ◆Increased risk of human error in payroll or scheduling
- ◆Poor employee experience
- ◆Compliance issues due to inconsistent record-keeping

→ Current Workarounds: Excel spreadsheets, disjointed HR software, paper-based records, manual compliance checks.

3. Proposed Solution

- Solution Description: A centralized, web-based Workforce Administration Platform that streamlines employee records, scheduling, compliance tracking, and integrates with existing HR and payroll tools.

4. Solution Validation

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Plan How will you validate this solution with real users?

- Prototype testing with 3 HR departments
- Usability testing of core features (scheduling, data entry)
- A/B test with current vs. new system for task completion time
- Feedback loop via in-app surveys and interviews

5. Expected Outcomes

What changes will success bring?

- 30–50% time saved in scheduling and data management
- Reduction in compliance risks due to built-in alerts and audits
- Improved employee satisfaction with clear, accessible records
- Better decision-making from real-time analytics

4.2 - Proposed System (Proposed Solution)

Key Components of the Proposed Solution

1. Employee Lifecycle Management

- Centralized employee profiles with personal, professional, and contractual data
- Automated onboarding and offboarding workflows
- Document management and digital signatures

2. Attendance & Time Tracking

- Real-time tracking via mobile app, biometric integration, or geolocation
- Shift scheduling and calendar sync
- Overtime calculation and approval workflows

3. Leave & Absence Management

- Self-service leave requests with policy-based approvals
- Leave balance tracking and holiday calendar
- Notifications and escalation alerts

4. Payroll Integration

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- Seamless sync with payroll systems for accurate compensation
- Automated handling of bonuses, deductions, and reimbursements
- Generation of payslips and tax documents

5. Compliance & Audit Readiness

- Built-in rules engine for labor law and policy enforcement
- Audit trails and version history for all changes
- Alerts for compliance violations and deadlines

6. Analytics & Reporting

- Dashboards for HR, managers, and leadership
- Workforce metrics: attendance, attrition, performance, engagement
- Exportable reports in multiple formats (PDF, Excel, etc.)

7. User Experience & Accessibility

- Role-based dashboards for HR, employees, and managers
- Mobile-friendly design and responsive UI
- Accessibility features for inclusive use

8. Integration & Extensibility

- RESTful APIs for ERP, payroll, and communication tools
- Webhooks for real-time event notifications
- Optional chatbot or voice assistant for HR queries

Technology Stack (Salesforce-Based)

- **Frontend:** Lightning Web Components (LWC)
- **Backend:** Apex, Triggers, Batch Apex
- **Database:** Salesforce Objects (Custom & Standard)
- **Integration:** REST/SOAP APIs, MuleSoft
- **Security:** OAuth 2.0, Salesforce Shield
- **Analytics:** Salesforce Dashboards, Einstein Analytics

4.3 - Solution Architecture

Architectural Overview

The architecture is modular, cloud-native, and built on Salesforce technologies, enabling seamless integration, automation, and extensibility. It includes:

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- **Presentation Layer** for user interaction
- **Application Layer** for business logic
- **Data Layer** for storage and retrieval
- **Integration Layer** for external system connectivity
- **Security Layer** for access control and compliance

☐ 1. Presentation Layer (User Interface)

- **Lightning Web Components (LWC)** for responsive, dynamic UI
- **Role-Based Dashboards** for HR, Managers, and Employees
- **Mobile Access** via Salesforce Mobile App or React Native (optional)
- **Self-Service Portals** for leave, attendance, and profile updates

☐ 2. Application Layer (Business Logic)

- **Apex Classes & Triggers** for core HR workflows (onboarding, leave approvals, etc.)
- **Process Builder / Flow** for declarative automation
- **Batch Apex / Queueable Apex** for asynchronous processing (e.g., payroll sync)
- **Validation Rules** to enforce data integrity and policy compliance

☐ 3. Data Layer (Storage & Management)

- **Standard & Custom Salesforce Objects:**
 - Employee
 - Attendance
 - Leave Request
 - Payroll Record
 - Compliance Log
- **Field-Level Security & Audit Trails** via Salesforce Shield
- **Document Storage** using Salesforce Files or external storage (e.g., SharePoint)

☐ 4. Integration Layer (Connectivity)

- **Salesforce REST/SOAP APIs** for external system communication
- **MuleSoft Anypoint Platform** for advanced integrations (ERP, payroll, HRIS)
- **Webhooks** for real-time event notifications
- **Email & Calendar Sync** for approvals and scheduling

☐ 5. Security & Compliance Layer

- **OAuth 2.0 / SSO** for secure authentication
- **Role-Based Access Control (RBAC)** for data visibility
- **Salesforce Shield** for field audit trails, encryption, and event monitoring
- **Compliance Engine** for labor law enforcement and reporting

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□ 6. Analytics & Reporting Layer

- **Salesforce Reports & Dashboards** for real-time insights
- **Einstein Analytics** for predictive modeling (e.g., attrition risk)
- **Export Tools** for generating audit-ready reports in PDF/Excel formats

□ 7. DevOps & Deployment

- **Salesforce DX** for source-driven development
- **GitHub / GitLab** for version control
- **CI/CD Pipelines** for automated testing and deployment
- **Change Sets / Unlocked Packages** for environment migration

Architecture Diagram (Conceptual Layout)

[User Interface]



[Lightning Web Components]



[Apex Logic / Flows / Triggers]



[Salesforce Objects (Employee, Leave, Attendance)]



[Integration Layer (MuleSoft, APIs, Webhooks)]



[External Systems (ERP, Payroll, HRIS)]

